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RESEARCH ARTICLE

Prevalence of Alzheimer's disease dementia in the 50 US states and 3142 counties: A population estimate using the 2020 bridged-race postcensal from the National Center for Health Statistics

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Abstract

INTRODUCTION: This study estimates the prevalence and number of people living with Alzheimer's disease (AD) dementia in 50 US states and 3142 counties.

METHODS: We used cognitive data from the Chicago Health and Aging Project, a population-based study, and combined it with the National Center for Health Statistics 2020 bridged-race population estimates to determine the prevalence of AD in adults ≥65 years.

RESULTS: A higher prevalence of AD was estimated in the east and southeastern regions of the United States, with the highest in Maryland (12.9%), New York (12.7%), and Mississippi (12.5%). US states with the highest number of people with AD were California, Florida, and Texas. Among larger counties, those with the highest prevalence of AD were Miami-Dade County in Florida, Baltimore city in Maryland, and Bronx County in New York.

DISCUSSION: The state- and county-specific estimates could help public health officials develop region-specific strategies for caring for people with AD.

KEYWORDS
2020 US prevalence, aging, Alzheimer's disease, dementia, epidemiology, prevalence

1 | INTRODUCTION

Alzheimer's disease (AD) and related dementias possess tremendous health, social, and economic burdens worldwide.[1] In the United States, the financial costs of caring for people with AD dementia are estimated at $321 billion (≈$50,000 per person) in 2022, including $206 billion in spending for Medicare and Medicaid.[2] Therefore, to better plan the financial costs of caring for people with AD dementia across the United States, it is necessary to provide state-specific estimates of the number of people with AD dementia, that is, the disease's prevalence. However, there are little to no surveillance systems to monitor the development of AD dementia. Medical records could be an alternative approach for estimating the prevalence of AD dementia in the population, albeit they are prone to misclassification due to measure errors[3–6] as well as individuals with cognitive impairment may avoid seeking professional care because of mental health stigma.[7] Therefore, novel approaches are required to estimate the state-specific prevalence and number of adults with AD dementia.

In our earlier work, we developed a dementia likelihood score to estimate the prevalence of AD dementia based on the data from a population-based study, that is, the Chicago Health and Aging Project (CHAP),[8] combined with US Census data.[9] Using this approach, we estimated the number of people with clinical AD dementia and mild cognitive impairment in the United States from 2020 to 2060.[9] In the current study, we will estimate the prevalence and number of people living with AD dementia in each of the 50 US states based on demographic characteristics in their respective counties. Additionally, we will provide the prevalence estimates for 3142 counties. These estimates could help public health programs better plan the budget for caring for people with AD dementia.

RESEARCH IN CONTEXT

1. **Systematic Review**: Estimating the prevalence of Alzheimer's disease (AD dementia at the county and state levels across the United States could help public health officials understand the burden (e.g., institutional care) of the disease.
2. **Interpretation**: We used demographic and cognitive data from a population-based study (i.e., Chicago Health and Aging Project) and combined it with the National Center for Health Statistics 2020 bridged-race population estimates and showed that the highest prevalence of AD dementia was estimated in the east and southeastern regions of the United States (i.e., Maryland, New York, and Mississippi) while states with the highest number of people with AD dementia were California, Florida, and Texas.
3. **Future Directions**: We used a single population study to estimate the role of demographics on AD prevalence; future research should use all population-based studies across the United States and re-evaluate the region-specific contribution of demographic factors to dementia risk while accounting for other risk factors.

2 | METHODS

2.1 | Chicago Health and Aging Project

CHAP is a prospective population-based cohort study designed to determine and evaluate the risk factors of AD dementia in the general population. Initiated in 1993, CHAP enrolled residents aged ≥65 years from a geographically defined community on the south side of Chicago. Of all individuals invited to participate in the study, 6157 (78.7% of all age-eligible people) responded to the invitation and enrolled. During the first study period (1993–2012), the cohort was extended with successive cohorts of residents reaching the age of 65. The study enrolled 10,802 people in the study catchment area. Data, including demographic and neurocognitive tests, were obtained via structured self- or interviewer-administered questionnaires during in-home assessments in 3-year cycles. A detailed description of the design and objectives has been reported previously.[10–12]

All participants provided informed consent to participate in the study. The Rush University Medical Center Institutional Review Board approved this study.

2.2 | Demographic characteristics

Race/ethnicity, sex, and education (years of formal schooling) were determined using the 1990 US census questions. Age was computed by subtracting the interview date from the date of birth.

2.3 | Neuropsychological testing and AD dementia

In CHAP, cognitive function was evaluated using a short battery of four neuropsychological tests, including two tests of episodic memory (immediate and delayed story recall),[13,14] a test of perceptual speed,[15] and a test of general orientation and global cognition (the Mini-Mental State Examination).[16] Using these short-battery test scores and demographic characteristics of 10,342 participants (96% of the total sample) with 36,408 cognitive assessments between 1993 and 2012, we determined the person-specific likelihood of AD dementia, as described earlier.[8] The participant-specific likelihood score is the probability that a study participant would be diagnosed with AD dementia at a given visit. The likelihood score ranges from 0 to 1, with 0 suggesting the lowest probability of being diagnosed with AD dementia and 1 indicating the highest chance. Details of how we developed the likelihood score have been described previously.[8]

2.4 | National Center for Health Statistics; 2020 bridged-race postcensal population estimates

The 2020 bridged-race postcensal estimates are produced by the Population Estimates Program of the US Census Bureau in collaboration with the National Center for Health Statistics (NCHS). These estimates result from bridging the 31 potential race categories collected in the 2010 census on race and ethnicity to four race categories: White, Black or African American, American Indian or Alaska Native, and Asian or Pacific Islander.[17] These population estimates are further divided by age group, sex, and ethnicity (Hispanic or Latino vs. not Hispanic or Latino). For our investigation, we focused on data for US residents ≥65 years to align with the age range of the CHAP population.

2.5 | Statistical analysis

To estimate the number of adults aged≥ 65 years with AD dementia by states in the United States, we went through a multi-step approach that uses (1) CHAP demographic and cognitive data as well as the (2) NCHS 2020 bridged-race population estimates. In the first step, we used a generalized additive quasibinomial regression model using data from the CHAP cohort. The generalized additive quasibinomial regression model allows us to include the person-specific likelihood score, which ranges from 0 to 1, as a dependent variable. The independent variables included age, sex, race/ethnicity, and education. In addition, the model included person-specific random effects for years since the first cognitive assessment and a baseline probability of AD dementia.[8] To apply the CHAP regression coefficients to the NCHS data and ultimately calculate the prevalence of AD dementia, we organized the categories of CHAP and NCHS in the same order. For example, we created 5-year age groups; that is such as 65 to 74, 75 to 79, 80 to 84, and ≥85 years. Sex includes men and women. Race/ethnicity comprised White, Black, and Hispanic individuals. Education is not available in NCHS, but given its strong effect on cognition, we adjusted our regression model by years of formal schooling. Therefore, our regression coefficients are controlled for education level. In the second step, we applied the regression coefficients from the CHAP model to the NCHS. Specifically, we multiplied the regression coefficients by the population estimates at the county level from the NCHS data for each group defined by combinations of age group, sex, and race/ethnicity; that is, $x \times \beta$, where x is the matrix of population estimates from the NCHS specific to each county and β is the vector of coefficient estimates from our model output using data from CHAP. The inverse logistic function ($exp[x \times \beta] / [1 + exp(x \times \beta)]$) is then applied to calculate the prevalence of AD dementia in each county. With the estimated prevalence data for each county in the United States, in the third step, we calculated the number of people with AD dementia by multiplying the prevalence by the total number of people aged ≥65 years in each county. Finally, to provide the prevalence of AD dementia by state, we sum the county-level estimates (i.e., the number of people with the disease) and divide by the total number of people aged ≥65 years respective to each state. Visualization of the statistical approach is shown in Figure S1 in supporting information.

3 | RESULTS

The demographic characteristics and their respective associations with the likelihood of AD dementia are shown in Table 1. In CHAP, individuals aged ≥85 years comprised 8% of the population, women 61%, Black individuals 63%, and Hispanic individuals 1.3%. The risk of AD dementia increases exponentially with age; compared to individuals in the 65 to 69 age range, those in the 70 to 74, 75 to 79, 80 to 84, and 85+ years had an odds ratio (OR) of 1.78 (95% confidence interval [CI] 1.64, 1.93), 3.08 (95% CI 2.85, 3.33), 6.05 (95% CI 5.59, 6.55), and 14.8 (95% CI 13.6, 16.0), respectively (Table 1). Women had a risk of AD dementia 1.13 times higher (OR 1.13 and 95% CI 1.08, 1.18) than men. Compared to Whites, Black and Hispanic individuals had a higher risk of AD dementia, ORs 2.50 (95% CI 2.37, 2.63) and 1.73 (95% CI 1.40, 2.13), respectively. Education, estimated as years of schooling, was associated with lower odds of AD dementia, respectively, for 1 standard deviation increase (3.5 years), the OR (95% CI) was 0.67 (0.66, 0.69).

3.1 | Prevalence of AD dementia by US states

We estimated the prevalence based on the estimates in Table 1 and the NCHS demographics data, such as age, sex, and race/ethnicity. Therefore, states with more people aged ≥85 years, women, and more minorities (especially Black individuals) will have a greater prevalence of AD dementia. Figure 1 shows the prevalence (Figure 1A), and the number of people (Figure 1B) with AD dementia by state across the United States. Table 2 shows the top 10 states with higher prevalence and number of people with AD dementia and their respective demographics. The highest prevalence of AD dementia is estimated in the east and southeastern regions of the United States, with the highest prevalence in Maryland (12.9%), New York (12.7%), Mississippi (12.5%), and Florida (12.5%). California and Illinois also have a higher estimated prevalence of 12%, respectively. The highest prevalence of AD in Maryland is explained by the greater number of people age ≥85 (12.2%), together with the higher proportion of Black individuals (25.4%). The top three states with the highest number of people with AD dementia are California, Florida, and Texas. The number of people with AD is primarily driven by the absolute number of people aged ≥65 years. For example, with about 5.9 million people aged ≥65 years and a prevalence of 12%, California has an estimated number of people with AD dementia of 720,000 (Table 2). Table S1 in supporting information shows the prevalence and number of people with AD dementia in the 50 US states.

3.2 | Prevalence of AD dementia by US counties

Figure 2 shows the prevalence (Figure 2A), and the number of people (Figure 2B) with AD dementia by county across the United States. Table 3 shows the top 10 counties—among those with a population of ≥65 years greater than 10,000—with higher prevalence and number of people with AD dementia and their respective demographics. Table S2 in supporting information shows the prevalence and number of people with AD dementia in 3142 counties of the 50 US states. Miami-Dade County in Florida, Baltimore city in Maryland, and Bronx County in New York have the highest prevalence of AD dementia, with 16.6% of people ≥65 years. A combination of specific demographic characteristics explains the higher prevalence. For example, Bronx County in New York comprises 14.0% of individuals aged ≥85; 30.1% Black and 46.9% Hispanic individuals. The counties with more individuals with AD dementia are Los Angeles County in California, with about 190,000 estimated cases of AD dementia from 1.4 million individuals aged ≥65 years, followed by Cook County in Illinois, with about 108,000 estimated cases from population of 792,000 individuals ≥65 years (Table 3).

4 | DISCUSSION

In this investigation, we used demographic and cognitive data from a population-based study (i.e., CHAP) and combined it with the NCHS 2020 bridged-race population estimates to determine the prevalence of AD dementia for adults ≥65 years in each of the 50 US states and their respective 3142 counties. The highest prevalence of AD dementia was estimated in the east and southeastern regions of the United

TABLE 1 Regression coefficients with odds ratios of demographic factors in the Chicago Health and Aging Project.

			Regression coefficients			
		Baseline characteristics	Beta	OR	95% CI	*P*-value
Age						
	65 to 69 years, *n* (%)	4460 (43.1)	0	1	reference	
	70 to 74 years, *n* (%)	2692 (26.0)	0.577	1.78	1.64, 1.93	<0.001
	75 to 79 years, *n* (%)	1439 (13.9)	1.126	3.08	2.85, 3.33	<0.001
	80 to 84 years, *n* (%)	931 (9.0)	1.8	6.05	5.59, 6.55	<0.001
	85 years or more, *n* (%)	818 (7.9)	2.693	14.78	13.63, 16.04	<0.001
Sex						
	Men, *n* (%)	3999 (38.7)	0	1	reference	
	Women, *n* (%)	6341 (61.3)	0.123	1.13	1.08, 1.18	<0.001
Race						
	White, *n* (%)	3742 (36.2)	0	1	reference	
	Black or African American, *n* (%)	6466 (62.5)	0.915	2.5	2.37, 2.63	<0.001
	Hispanic, *n* (%)	132 (1.3)	0.548	1.73	1.4, 2.13	<0.001
Education						
	Years of schooling, mean (SD)	12.3 (3.5)	-0.398	0.67	0.66, 0.69	<0.001

Note: A generalized additive quasibinomial regression model with person-specific likelihood score (range 0 to 1) as dependent variable and age, sex, race/ethnicity, and education as independent variables. Estimate and OR for education is for 1 standard deviation increase. Intercept of the model is −3.455.
Abbreviations: CI, confidence interval; OR, odds ratio; SD, standard deviation.

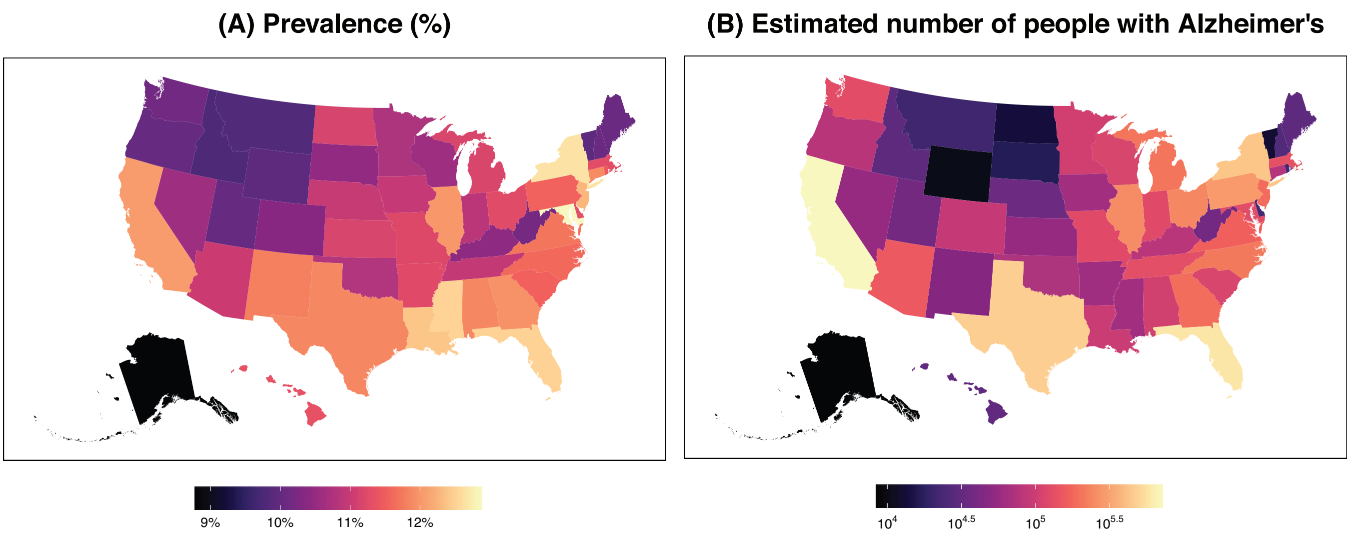

FIGURE 1 Estimated prevalence and the number of people with Alzheimer's disease dementia in 50 US states for adults ≥65 years.

States, with the highest prevalence in Maryland, New York, Mississippi, and Florida. California and Illinois were also among the 10 states with a higher prevalence of AD. US states with the highest number of people with AD dementia were California, Texas, and Florida. The number of people with AD is primarily driven by the absolute number of people ≥65 years. These estimates could help public health officials to understand the burden of disease (e.g., demand for caregiver counseling and institutional care) at the county and state levels and develop adequate strategies for identifying and caring for people with AD dementia.

Most studies on the prevalence of AD dementia are based on data from medical reports, claims, and nationally representative surveys.[3,6,18–24] Both claims and surveys provide valuable data; however, recent reports have shown that only half of the study participants were diagnosed by both measures, emphasizing the

TABLE 2 US states with the highest estimated prevalence or number of people with AD dementia among adults ≥65 years.

States	No. of people age 65 years and older, in thousands	Calculated estimates: Prevalence, % (95% CI)	Calculated estimates: No. (95% CI) of people with AD, in thousands	2020 demographics: Age 85 years or more, %	2020 demographics: Women, %	2020 demographics: Black or African American, %	2020 demographics: Hispanic, %
States with highest prevalence of AD dementia							
Maryland	987.4	12.9 (12.2, 13.5)	127.2 (120.9, 133.5)	12.2	56.7	25.4	3.7
New York	3369.6	12.7 (11.9, 13.4)	426.5 (400.2, 452.7)	13.7	56.7	12.3	11.7
Mississippi	500.0	12.5 (11.9, 13.1)	62.5 (59.6, 65.5)	10.7	56.4	27.9	1.2
Florida	4638.1	12.5 (11.6, 13.4)	579.9 (539.9, 620.0)	12.9	54.9	9.1	15.7
Louisiana	763.8	12.4 (11.8, 13.0)	94.7 (90.0, 99.4)	10.9	56.0	25.3	2.8
New Jersey	1510.0	12.3 (11.5, 13.0)	185.3 (174.0, 196.7)	13.3	56.6	10.3	10.9
California	5976.2	12.0 (11.1, 13.0)	719.7 (665.0, 774.4)	12.7	55.4	5.4	21.1
Illinois	2089.2	12.0 (11.3, 12.7)	250.6 (236.4, 264.8)	12.8	56.1	11.8	7.4
Georgia	1574.7	12.0 (11.4, 12.6)	188.3 (178.7, 197.8)	9.9	56.4	24.5	3.1
Connecticut	646.0	11.9 (11.2, 12.6)	76.8 (72.5, 81.1)	14.0	56.1	7.1	6.7
States with highest no. of people with AD dementia							
California	5976.2	12.0 (11.1, 13.0)	719.7 (665, 774.4)	12.7	55.4	5.4	21.1
Florida	4638.1	12.5 (11.6, 13.4)	579.9 (539.9, 620.0)	12.9	54.9	9.1	15.7
Texas	3873.7	11.9 (10.9, 12.8)	459.3 (422.7, 496.0)	10.7	55.2	9.6	23.7
New York	3369.6	12.7 (11.9, 13.4)	426.5 (400.2, 452.7)	13.7	56.7	12.3	11.7
Pennsylvania	2447.7	11.5 (10.9, 12.1)	282.1 (267.7, 296.5)	13.4	55.9	7.4	2.6
Illinois	2089.2	12.0 (11.3, 12.7)	250.6 (236.4, 264.8)	12.8	56.1	11.8	7.4
Ohio	2097.6	11.3 (10.7, 11.8)	236.2 (224.4, 248.0)	12.2	55.8	9.3	1.4
North Carolina	1814.5	11.6 (11.0, 12.2)	210.5 (199.9, 221.1)	10.7	56.2	17.4	2.4
Michigan	1812.4	11.2 (10.6, 11.8)	202.8 (192.5, 213.1)	11.7	55.1	10.4	1.9
Georgia	1574.7	12.0 (11.4, 12.6)	188.3 (178.7, 197.8)	9.9	56.4	24.5	3.1

Note: The top 10 states with the highest prevalence or number of people with AD dementia are shown in Table 2. Table S1 in supporting information shows the prevalence and number of people with AD dementia in 50 states of the United States.
Abbreviations: AD, Alzheimer's disease; CI, confidence interval; No., number.

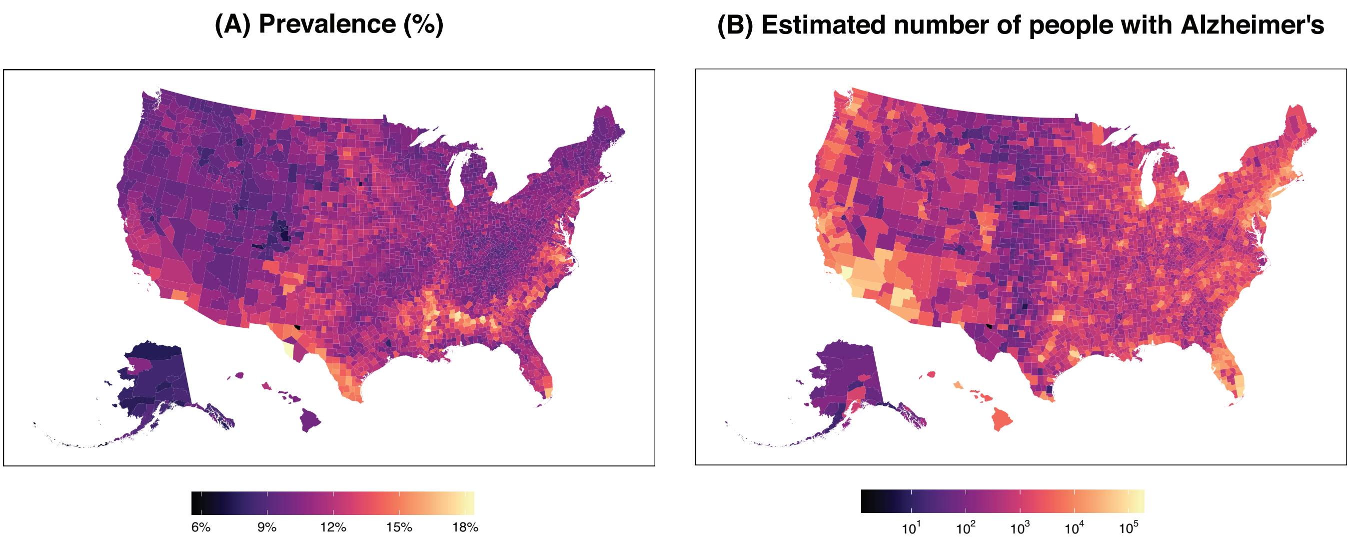

FIGURE 2 Estimated prevalence and the number of people with Alzheimer's disease dementia in 3142 US counties for adults 65 years.

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TABLE 3 US counties with the highest estimated prevalence or number of people with AD dementia among adults ≥65 years

		Calculated estimates		2020 demographics			
Counties	No. of people age 65 years and older, in thousands	Prevalence, % (95% CI)	No. (95% CI) of people with AD, in thousands	Age 85 years or more, %	Women, %	Black or African American, %	Hispanic, %
Counties with highest prevalence of AD dementia							
Miami-Dade County, Florida	459.9	16.6 (14.6, 18.7)	76.6 (67.3, 85.9)	14.1	58.4	12.8	70.4
Baltimore city, Maryland	87.8	16.6 (15.9, 17.3)	14.6 (14, 15.2)	12.0	59.6	64.9	1.8
Bronx County, New York	192.6	16.6 (15.1, 18.1)	31.9 (29, 34.8)	14.0	60.3	30.1	46.9
Prince George's County, Maryland	129.9	16.1 (15.3, 16.8)	20.8 (19.9, 21.8)	9.8	58.7	68.4	6.0
Hinds County, Mississippi	35.3	15.5 (14.8, 16.1)	5.5 (5.2, 5.7)	11.4	58.5	59.7	0.3
Orleans Parish, Louisiana	63.2	15.4 (14.7, 16.1)	9.7 (9.3, 10.2)	10.7	56.9	59.3	3.4
Dougherty County, Georgia	14.7	15.3 (14.7, 16.0)	2.3 (2.2, 2.4)	11.7	58.4	55.9	1.1
Orangeburg County, South Carolina	18.0	15.2 (14.5, 15.8)	2.7 (2.6, 2.8)	10.8	57.0	52.9	0.9
Imperial County, California	24.5	15.0 (13.1, 17.0)	3.7 (3.2, 4.2)	13.4	55.2	1.5	74.7
El Paso County, Texas	107.9	15.0 (13.0, 17.1)	16.2 (14, 18.4)	12.6	57.2	2.2	78.6
Counties with highest no. of people with AD dementia							
Los Angeles County, California	1444.5	13.2 (12.1, 14.3)	190.3 (174.2, 206.3)	13.4	56.4	8.8	30.8
Cook County, Illinois	792.1	13.6 (12.8, 14.4)	107.6 (101.2, 114)	13.1	57.5	23.4	13.1
Maricopa County, Arizona	729.8	11.1 (10.4, 11.8)	81.0 (75.7, 86.4)	11.5	54.9	3.5	12.0
Miami-Dade County, Florida	459.9	16.6 (14.6, 18.7)	76.6 (67.3, 85.9)	14.1	58.4	12.8	70.4
Harris County, Texas	532.1	12.2 (11.2, 13.1)	64.8 (59.8, 69.8)	9.9	55.5	18.0	24.3
San Diego County, California	496.4	11.8 (10.9, 12.6)	58.4 (54.1, 62.7)	13.0	55.3	3.7	18.1
Orange County, California	497.7	11.6 (10.8, 12.4)	57.8 (53.7, 61.9)	13.1	55.5	1.4	16.0
Kings County, New York	376.4	15.0 (14.1, 15.9)	56.5 (53.2, 59.9)	13.7	58.9	33.4	14.9
Queens County, New York	377.3	13.7 (12.8, 14.6)	51.7 (48.1, 55.2)	13.7	57.3	18.0	19.6
Palm Beach County, Florida	374.6	13.8 (13.0, 14.6)	51.6 (48.6, 54.6)	17.4	55.6	9.3	10.7

Notes: Table 3 shows the top 10 counties (which had a population of 10,000 or more individuals aged 65 years and older) with the highest prevalence or number of people with AD dementia. Table S2 in supporting information shows the prevalence and number of people with AD dementia in 3142 counties of the 50 US States.
Abbreviations: AD, Alzheimer's disease; CI, confidence interval; No., number.

discrepancy between methods and the challenge of identifying people with cognitive impairment in the general population.[3–6] Specifically, most disagreements between claims and survey data were among minorities, including Black and Hispanic individuals.[6] With the increasing proportion of minorities in the United States, the discrepancies between claim- and survey-based data for the diagnosis of dementia are expected to increase, ultimately underestimating the prevalence of dementia in minorities. Consequently, in the long term, resources for individuals living with dementia in communities with higher proportions of minorities might not meet the disease burden. In our study, we provide an alternative approach for measuring the prevalence and number of people with AD dementia in the United States.[9] Our methods focus on two data sources: (1) a population-based study composed of a large number of minorities (i.e., Black individuals) to ensure adequate power for estimating the effect of the race on the risk of A dementia; (2) census-based population estimates to account for the distribution of age, sex, and racial–ethnic groups at the county level across the United States. Using this approach, we have published the overall prevalence of AD dementia in the United States,[9] and now we provided the prevalence and number of people living with AD dementia in each of the 50 states and 3142 counties of the United States.

We estimated the prevalence of AD dementia across the United States based on the state-specific distribution of demographic factors, such as age, sex, and race/ethnicity. Therefore, states with more people aged ≥85 years, women, and minorities have a greater prevalence of AD dementia. However, in addition to these critical demographic factors, there are other risk factors, such as cardiovascular risk factors (e.g., diabetes, hypertension)[25] and lifestyle factors (e.g., diet, cognitive activity),[12,26,27] that contribute to the risk of AD dementia and, ultimately to the prevalence. Such data are unavailable at the county level, and we cannot incorporate them into our estimates. In addition, another characteristic of our approach is that the effect estimates of demographic factors on dementia risk came from a single population-based cohort study (i.e., CHAP) and applied to 3142 counties across the United States. Although the relationship between demographic factors and dementia should be similar across populations (e.g., older people are at high risk of dementia), the effect of other risk factors (e.g., lifestyle, diabetes, hypertension) on dementia risk may attenuate or amplify the role of demographic factors on AD prevalence.[12,25–27] For example, for states that rank higher in health determinants and outcomes (i.e., America's Health Rankings),[28] the contribution of demographic factors, such as older age, to AD prevalence will be smaller compared to states with similar demographics but less optimal health determinants and outcomes (e.g., scoring lower by America's Health Rankings). To overcome such limitations, we propose that future investigations on the prevalence of AD should identify population-based studies across the United States and re-evaluate the contribution of demographic factors to dementia risk while accounting for other risk factors (e.g., lifestyle factors). Then, these region-specific AD dementia estimates should be applied to counties and state census data in which the population-based study was conducted. For example, estimates of CHAP should be applied to the Midwest census region, while results from the Einstein Aging Study and Framingham Heart Study should be applied to the East census region. Such research is very much needed to understand the disease heterogeneity and risk factor distribution across the US counties and states to design better preventive strategies and programs. Another limitation is that our study focused on three major racial/ethnic groups in the U.S. (i.e., Black/African American, Hispanic, and White) and assumed that other races (i.e., Asian American and American Indian/Alaska Native individuals) have a similar risk to White individuals. However, according to a recent study investigating inequalities in dementia incidence across six racial and ethnic groups, American Indian/Alaska Native had a higher incidence compared to White individuals, while Asian American individuals had the lowest.[29] In addition, the Hispanic population within the CHAP study is relatively small compared to the Black and White populations. We adjusted for education level in our regression model, given that it is a strong predictor of cognitive impairment. However, because education is not available in 2020 bridged-race postcensal from the NCHS, we used an average of 12 years for all states, which also might be different by the states and counties. The strengths of our study include the population of a large proportion of Black individuals, powering the study to examine the role of demographic factors on dementia risk among Black individuals, as well as an 18-year prospective design and repeated uniform cognitive assessments.[10]

5 | CONCLUSION

In conclusion, using the demographic and cognitive data from a population-based study (i.e., CHAP) and combining it with the NCHS 2020 bridged-race population estimates, we showed that the prevalence of AD dementia was heterogeneous across states in the United States with the highest rates in the east and southeast regions. We also concluded that most people living with AD dementia are in states with a larger population aged ≥65 years, including California, Texas, and Florida. Our county-specific estimates on the prevalence and number of people with AD dementia could be helpful to public health officials to plan better the budget for caring for people with AD dementia.

AUTHOR CONTRIBUTIONS

Klodian Dhana and Kumar B. Rajan designed and conceptualized the study, conducted data analysis, interpreted the findings, and drafted and revised the manuscript for intellectual content. Todd Beck conducted data analysis, and reviewed and revised the manuscript. Pankaja Desai and Robert S. Wilson reviewed and revised the manuscript and provided intellectual content. Denis A. Evans oversaw data collection and supervised the study.

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CONFLICT OF INTEREST STATEMENT

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DATA AVAILABILITY STATEMENT

De-identified data may be available on request for qualified investigators from www.riha.rush.edu/dataportal.html

CONSENT STATEMENT

The institutional review board of the Rush University Medical Center approved the CHAP study protocols, and all participants provided written consent for population interviews, blood collection, and clinical evaluations.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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